

1 Introduction - Motivation

According to Census data, relocation in America is at its lowest rate on record (Tavernise, 2019). This is concerning because the lack of people moving for work could negatively impact the economy (Kopf, 2018). Still, trying to find the right place to live can be difficult.

Today, a few popular websites to help people find the “best places to live”. Livability and Teleport provide a profile of cities and a breakdown of their basic statistics. Niche and AreaVibes show neighborhoods and their “livability” scores, which are calculated based on several categories (e.g., cost, crime, etc.). These individual groupings also receive a rating, and all scores contribute to a locale’s ranking in comparison to nearby areas. Finally, the data presentations are static and arranged in lists.

There are several limits to the current practice. First, none of the sites described above offer a list of neighborhoods similar to the one a user input. Next, while searches are filterable, there are no personalized recommendations for a user. Moreover, these sites do not have information about local grocery stores or other essential business in an area, which is important when deciding on a new residence. Lastly, aside from home sale maps, none of the sites offer an interactive visualization.

For our project, we are developing a tool that will make the process of relocation easier. Our goal is to reduce the time and stress of searching for a new location to live by providing people with useful and personalized information on neighborhoods from across the country. We are confident that our tool will encourage more mobility because it will give people the confidence to potentially relocate when they might have been hesitant to before.

2 Problem Definition

Our team is creating a tool to help people find the right neighborhood. However, individuals might not know what factors are truly most important to them. So, we believe the best approach is to provide people with useful information on new areas based on locales they already know and love. We plan to utilize a user’s input of an address, determine its neighborhood, and present a list of recommended locations from across the country that are like the user’s identified neighborhood. The final output is an interactive visualization that showcases the recommended neighborhoods based on a user’s location inputs.

3 Literature Survey

First, the GARM chapter provided important details on Census tracts (U.S. Census Bureau, 1994). They are often used as proxies for neighborhoods, which our group exploited. Next, several papers looked at neighborhood factors people find important, which our group considered. Okulicz-Kozaryn (2011) compared livability and satisfaction of locations and finds only a .36 correlation. Though, there may have been a bias towards subjective metrics. Our team treated all selected attributes equally. Ayub et al. (2020) examined housing indicators to better understand housing markets, but the focus was strictly on urban markets. Our dataset contains Census tracts from every type of locale. He and Xia (2020) determined rental effects for a housing market are largely based on amenities. Yet, their analysis only focused on Barcelona, while we highlight the entire US. Azadeh and Moghaddam (2012) showed economic and government factors can predict fluctuations in housing markets, but they did not consider other factors. We incorporated as many neighborhood attributes as possible. Ashkexari-Toussi et al. (2019) assessed sentiment in geotagged photos, but they used old data. We retrieved the most recent data available.

A key component of our project will be to utilize a cluster model to group similar neighborhoods. Fahad et al. (2014) evaluates well-known single clustering algorithms. However, there is not a single algorithm that performs best on all datasets. Cluster ensembles can overcome this issue by combining different clustering outputs into one consensus result, and they are surveyed in Boongoen and Iam-On (2018). Finally, Halkidi et al. (2001), Liu et al. (2010), Rendón et al. (2011), and Arbelatriz et al. (2013) all reviewed several internal clustering measures, and they investigated and compared the validation properties of the indexes. Internal validity measures will be utilized for our evaluations and experiments.

In addition, we reviewed ranking and recommendation models as other possible approaches. Kiseleva et al. (2015) showed that a Naïve Bayes ranking on features for location recommendations performed better than other rankings, but historical data was not leveraged. Our intent is to allow users to input the locations they have lived. Liu and Yang (2015) improved VSRank recommendation by focusing on

negative similarity and people's preferences. Unfortunately, geography was not considered, but this is one of our key factors. Kim et al. (2020) improved collaborative filtering-based recommendations with sentiment analysis, but it could be improved by specialized sentiment rather than dictionary-based analysis. Su et al. (2020) created a vector of multiple similarities, which our project could utilize. Though, only MAE and RMSE were used to measure accuracy. We will use other types of metrics for our evaluations. Bobadilla et al. (2013) reviewed several recommendation algorithms, but none of them were as effective with sparse data or cold starts. To overcome this problem, we tried to reduce the sparsity in our final dataset. Renjith et al. (2020) examined personalized recommender systems on multiple criteria, which we aim to employ. Yet, their focus was exclusively on travel locations, while we collected more granular data. Overall, performing a ranking of the clustering results is feasible, but we do not believe a recommendation algorithm can be adequately deployed within the timeframe of the project.

Moreover, Xiaoyu et al. (2017), Al-Ghossein et al. (2018), and Wasid and Ali (2018) all employed a two-stage architecture. The first paper used matching and ranking on geotagged photos to recommend attractions, yet there was a large assumption about people's photo habits. For our project, we did not make any assumptions about human behavior. The second paper recommended hotels using clustering and ranking. Though, there may have been issues with the contextual data. Our dataset consists exclusively of numerical columns. The third paper used a two-stage approach to cluster then produce recommendations for movies. Unfortunately, the distance measure they employed was used specifically for their dataset, while we have tried to produce a more generalized model. Based on these three papers, we may utilize a two-stage approach of clustering and ranking in our final visualization tool.

Finally, we looked at how various models were applied to geographical problems, which is the basis of our project. Shashank and Schuurman (2019) showed how variable selection impacts walkability across different models, but the variables could have been combined to create a composite model that was more informative. With our model, all significant factors will be considered simultaneously. Nicholson et al. (2019) used spatial regressions on census blocks to estimate disaster vulnerability. The analysis demonstrated a feasible use case of neighborhood data, but platform selection may have produced bias. Our project includes data from all Census tracts to prevent bias. Sadler et al. (2019) scored areas and weighted expert opinions to improve a "healthfulness index". With our project, the data pertaining to neighborhood amenities have been weighted based on their distance from Census tract centers.

4 Proposed Method - Intuition and Description of Approaches

Our project exhibits several innovations that make it better than "state of the art". To start, we have enhanced the underlying data. Metrics that are typically used by the existing websites (e.g., income, demographics, etc.) have been combined with data on essential businesses, like grocery stores and healthcare centers. So, users will have an awareness of the amenities accessible in an area.

Next, our visualization tool will use the results of a clustering model that will be applied to Census tracts from all 50 states. The clustering algorithm will group similar tracts together based on the features in the improved dataset. So, when a user inputs an address into the application, the Census tract, or "neighborhood", of the address will be identified, and similar "neighborhoods" from across the US will be displayed for the user to review. This group of similar "neighborhoods" represents the initial step in generating personalized recommendations, which is not an available option on the existing websites.

Moreover, the user experience will be enriched with our visualization tool, "RELO". A major improvement is that the tool is exclusively focused on finding the best place for someone to live based on their input without limiting to a certain region of the US. RELO displays all potential relocation areas and provides users with the best possible results. Furthermore, the interface improves on competitors by making the style more welcoming and reducing the amount of excess information. Finally, RELO provides an innovative and dynamic map that users can zoom once a search result is selected.

4.1 Data Collection, Cleaning, and Preparation

The Census tract data comes from the Census 5-year American Community Survey Responses for 2018, which was found through tables available on data.census.gov. In total, eight Census tables were identified that contained data relevant to our project. After combining the data, there were around 74,000 Census tracts and 1,300 fields. The data was transformed to the true percentage of each population group.

Columns were dropped due to sparsity, high correlation with a more applicable column, or being too specific of a category. To prevent skewed results, extreme outliers were either truncated or the columns were dropped entirely. Null values were imputed, but rows with majority null values were dropped.

A Python script pulled the data on neighborhood amenities by looping through US zip codes and calling the Foursquare API, which has a high free data limit. The data collected for specific amenities was between 50,000-75,000 rows per amenity with several features. For each amenity, the data was combined into a CSV file, and OpenRefine was used for cleaning. Duplicates and venues that did not match the amenity type were dropped, places with similar names were merged, and incorrect tags were modified. Our team decided to first obtain data on grocery stores, but are currently adding other amenities.

4.3 Data Integration

The amenities data was aggregated and integrated with the Census data. For each Census tract, the number of venues of a specific amenity type were counted within 2, 5, 10, 25, and 50 miles of the center and placed into distinct columns. The location counts in these columns were weighted by the inverse of their distance to the center. To calculate the distance, the following algorithm was applied: 1) project the coordinate system onto a 2D plane; 2) build a KDTree of all the amenity's points; 3) iterate through Census tracts and use a tract's center as a query to the KDTree, along with the distance limit.

4.4 Analysis

All features in the final dataset are numeric. For the algorithms in our analysis to give equal weight to all features, the data will be scaled, which will improve the efficacy of our algorithms. Another important data transformation is dimensionality reduction, which further reduces the number of features. This technique helps address the “curse of dimensionality” that often occurs in high-dimensional spaces. Principal Component Analysis (PCA) will be used to transform the input data by projecting it into a lower number of dimensions called components. The components capture the variability of the input data through a linear combination of the features (Abdi and Williams, 2010).

Several clustering techniques will be implemented on the processed data. The discussion of which algorithms will be tested, how their performances will be evaluated, and how the final method will be chosen can be found in section “5 Experiments / Evaluation”. Once the optimal model has been chosen, it will be applied on the processed data to obtain a set of cluster assignments, which will be combined with the original preprocessed data. Thus, we will have an overall clustering of all the Census tracts in which tracts in the same group are more similar to one another than tracts from different groups. If time permits, we will try to produce multiple clustering results for each notable category in our dataset.

4.5 Visualization

The prototype of our visualization application has been created with Python Flask, and full interaction with the tool occurs in a web browser. Users begin by entering an address of a location they like, and the application will identify the corresponding Census tract by interacting with the Census Bureau's “Geocoding” API. Once this Census tract is captured, the final clustering results will be filtered based on the cluster assignment of the identified tract. The visualization tool will then display a map of the like Census tracts, or “neighborhoods”, in the same group as the identified tract, which users can consider as possible relocation options. In addition, the returned Census tracts will be ranked according to their Euclidean distance in the feature space from the identified tract, and a table of the top ranked locations will be displayed. Finally, the user may select any of the locations in this table, which will zoom the country map and display some of the amenities offered in the area. If there are no like “neighborhoods”, then instructions are shown to input another address. If time permits, we will try to give users the ability to search for similar neighborhoods based on a specific category, like average income.

5 Experiments / Evaluation

There are four main categories of clustering approaches: partitioning, hierarchical, density-based, and grid-based. Our group will evaluate several clustering methods to obtain the best results for our tool. Thus far, we are planning to evaluate the K-Means algorithm, which is a partitioning method. Additional models will be considered, including algorithms from the other categories and cluster ensembles. We are finalizing the pipeline to evaluate these different methods. Nonetheless, we have completed the design of our evaluations that will help us determine the best model to utilize.

5.1 Design of Upcoming Experiments / Evaluation

Unfortunately, the evaluation of clustering is difficult because there is no goal model or a set of true class labels for comparison. Since the true result is unknown, then we need to evaluate the models in other ways. So, we have determined a list of questions that our evaluations are designed to answer.

5.1.1 Does a non-random structure exist in the dataset?

Clustering results are only meaningful when there is non-random structure in the data. We need to assess the feasibility of clustering analysis on our cleaned dataset. So, we will employ a statistical method that utilizes the Hopkins statistic (Lawson and Jurs, 1990). It measures the probability that a given data set is generated by a uniform distribution (Han et al., 2012). The data has statistically significant clusters if the statistic is close to 0, but it is unlikely to have statistically significant clusters for values close to 0.5. If the Hopkins statistic is close to 1, then the test is inconclusive.

5.1.2 How good are the resulting clusters?

We will need to measure the quality of the clusterings generated by the different methods and compare them. Since ground truth labels are not available, we will rely on intrinsic criteria, which focus on the quality of the partition. These metrics evaluate the “goodness” of a clustering by considering the two properties expected in a first-rate clustering: compact and well separated groups (Han et al., 2012).

Over 30 intrinsic criteria can be found in the literature, and Arbelatriz et al. (2013) exhaustively tested many of them. Some measures had significantly better performance than others, but the Silhouette, Davies-Bouldin, and Calinski-Harabasz indexes performed well in a wide range of situations. So, we plan to use at least these three metrics during our evaluations and comparisons of the different clusterings. The chosen intrinsic indexes will be calculated for the clusterings produced by each of the methods we evaluate, and the method with the best overall measures will be chosen as our final model.

5.1.3 What is the “right” number of clusters in the dataset?

Some of the algorithms we will be evaluating, such as K-Means, require the number of clusters in a dataset be used as a parameter. The procedure of obtaining the correct number of clusters is also used to compare the characteristics of different size clusterings (Han et al., 2012). One popular approach is a heuristic known as the “elbow method”, in which the “elbow” in the curve of an internal validity measure (e.g., the Silhouette Coefficient) with respect to the number of clusters is determined.

A powerful statistical method, known as the “gap statistic”, can also be utilized. The approach consists of computing different clusterings of the data, increasing in the number of clusters, and comparing them to clusters of data generated with a uniform distribution. The inter-class distance matrix, Sw , is computed for both sets and compared. The correct number of clusters is where the widest gap appears between the Sw of the data and the uniform data (Tibshirani et al., 2002).

5.1.4 What are the features that make each cluster unique?

We will select a subset of the features that are important for identifying the differences between clusters. So, we will aggregate the data based on the generated clusters and retrieve the mean value for each variable. Then, we will calculate the variance of means between clusters within each variable and select the top variables with the highest variance.

5.2 Details of the Experiments / Evaluations

Please note that the results of the evaluations described above are still to be determined.

6 Plan of activities and distribution of team effort

Our team’s initial plan of activities, time estimates, and team member assignments were: data retrieval, organization, and cleaning (2 weeks: all members); model construction and validation (2 weeks: Bell, Janicki, Meyer, & Smootha); and development of interactive visualization (2 weeks: Bodie & Dassanayake). Most importantly, all team members have contributed similar amount of effort.

6.1 Revised plan of activities

Our team’s revised plan of activities, time estimates, and team member assignments are: production of the final dataset (1 week: Bell, Meyer, & Smootha); evaluation of clustering algorithms, selection of the final model, and generation of the results (2 weeks: Bell, Meyer, & Smootha); and generation of results ranking and completion of the interactive visualization (3 weeks: Bodie, Dassanayake, & Janicki).

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